

RESEARCH INTERESTS

Robotics | Safety-Critical Control Systems | Autonomous Systems | Motion Planning & Navigation

EDUCATION

Degree	Institute	CGPA	Year of Graduation
Master of Science	University of Michigan, Ann Arbor	4.0/4.0 (current)	2027 (expected)
Bachelor of Technology	Indian Institute of Technology, Kharagpur	8.77/10	2025

TECHNICAL SKILLS

- **Languages:** Python | MATLAB | C | Arduino | Visual Basic | LaTeX
- **Software:** ROS | Gazebo | RViz | MATLAB/Simulink | Fusion 360 | Pix4Dmapper

EXPERIENCE

- **Graduate Research | Distributed Autonomous Systems and Control Lab | Prof. Dimitra Panagou** *September 2025 - Present*
 - Implementing and analyzing safety filters: **backup control barrier functions**, **model predictive shielding**, and the **gatekeeper**
 - Investigating the core theoretical difference between these three safety filters on pre-defined **backup** and **nominal** controllers
 - Designed and simulated a series of scenarios to demonstrate the theoretical differences and the failure modes of each filter
- **Undergraduate Research | Unmanned Aerial Vehicles Lab | Prof. Sikha Hota** *August 2024 - July 2025*
 - Implemented **Biased PNG**, **Two-Stage PNG**, and **Time-to-go Polynomial Guidance** laws for 2D missile interception scenarios
 - Validated that the guidance algorithms achieved **desired impact angles** while satisfying **acceleration and FOV constraints**
 - Adapted **multi-planar** guidance laws for **3D interception** and worked to eliminate lateral acceleration profile discontinuities
 - Designed a **2-planar 3-Stage 3D Impact Angle Control Guidance (IACG)** law that can achieve any terminal impact direction
- **Research Intern | Control Law Group | RWR&DC | Hindustan Aeronautics Ltd** *May 2024 - July 2024*
 - Analyzed, in-depth, the **AFCS** (Automatic Flight Control System) and actuator mechanisms of a semi-autonomous helicopter
 - Integrated **Pitt-Peters** dynamic inflow model into the helicopter dynamics simulator to improve transient response predictions
 - Validated simulation results against flight test data, for **collective dump** input, identifying areas for future developments

PROJECTS

- **Academic Project | Model Predictive Control | Prof. Ilya Kolmanovsky** *Jan 2026 - Present*
 - Engineered a **constrained Model Predictive Controller** to achieve offset-free yaw rate tracking for vehicle lateral dynamics
 - Developed a **quadratic programming** solver utilizing a **dual projected gradient** algorithm for real-time MPC optimization
 - Integrated soft constraints via **slack variables** to ensure robust control performance against simulated plant-model mismatches
- **Academic Project | Non Linear Control of an Inverted Pendulum | Prof. Inigo Incer** *Jan 2026 - Present*
 - Engineered a full-state feedback controller for a **nonlinear underactuated system** using dimensionless state transformations
 - Implemented a **Luenberger observer**-based dynamic compensator to estimate velocities and to verify closed-loop stability
- **Unitree G2 High-Speed Trajectory Controller | Dimensional OS (Open Source)** *Feb 2026*
 - Upgraded the **control architecture** from a 10Hz reactive P/PD model to a 100Hz one to enable high-speed **trajectory tracking**
 - Implemented an **adaptive Pure Pursuit** controller to track the trajectory and velocities, generated by a novel **velocity profiler**
 - Incorporated a **PID Cross-Track controller** to eliminate lateral drifts and an adapter for **ROS** Twist message conversion
- **Academic Project | Online Learning for Control | Prof. Vasileios Tzoumas** *Sep 2025 - Dec 2025*
 - Implemented an **adaptive SSI-MPC** framework in **ROS2** to handle real-time wind disturbances using Recursive Least Squares
 - Deployed a **self-adaptive MPC** using expert predictors to execute **trajectory tracking** of UAVs with unknown flight dynamics
 - Integrated **Acados-based** MPC solvers within a **Gazebo SITL** for high-speed, autonomous multi-drone trajectory tracking
- **Academic Project | Aerospace Navigation and Guidance | Prof. Giusy Falcone** *Sep 2025 - Dec 2025*
 - Engineered a discrete-time **Kalman Filter** for accurate **state estimation** under a **Gaussian** process and measurement noise
 - Designed a computationally efficient **pursuit guidance law** for dynamic target interception in a high-speed engagement case
- **Aerial 2D Mapping and Estimation of Forest Area Cover | Prof. Sandeep Saha** *Jan 2024 - Apr 2024*
 - Automated **semantic segmentation** and feature extraction using a **U-Net CNN** to aid large-scale **environmental perception**
 - Processed aerial imagery with GPS and **altitude telemetry** to construct orthorectified 2D spatial maps using **Pix4Dmapper**
 - Enhanced spatial map resolution using **EDSR**, generating **Digital Surface Models** to estimate terrain elevation from samples
- **Aerial Carbon Emission Monitoring System | Prof. Srinibas Karmakar** *Aug 2023 - Nov 2023*
 - Designed a drone-deployed embedded payload with **MQ7** and **MQ135** sensors to execute real-time **telemetry data** acquisition
 - Developed an **ascent time trajectory algorithm** for an aerial platform setup by adopting a dynamic atmospheric state model
 - Engineered an altitude **state estimation** system using **BMP280** sensors and employed **nRF24L01** modules for **RF** telemetry

PUBLICATIONS

1. N. K. Singh, S. Pal, **A. D. Menon**, and S. Hota, “Impact Angle Constrained Guidance Laws for Stationary Target Interception in 3D Space,” in *AIAA SCITECH 2026 Forum*, 2026, p. 2348.

RELEVANT COURSEWORK

- **Control & Robotics:** Model Predictive Control | Online Learning for Control | Motion Planning and Control | Non-Linear Systems and Control | Embedded Control Systems | Robotics | Systems Laboratory
- **Dynamics & Guidance:** Aerospace Navigation and Guidance | Space Dynamics | Intermediate Dynamics | Flight Mechanics
- **Flight Controls:** Flight Vehicle Controls | Flight Stability & Control | Automatic Control of Aircraft | Helicopter Engineering